



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KPS FUEL AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

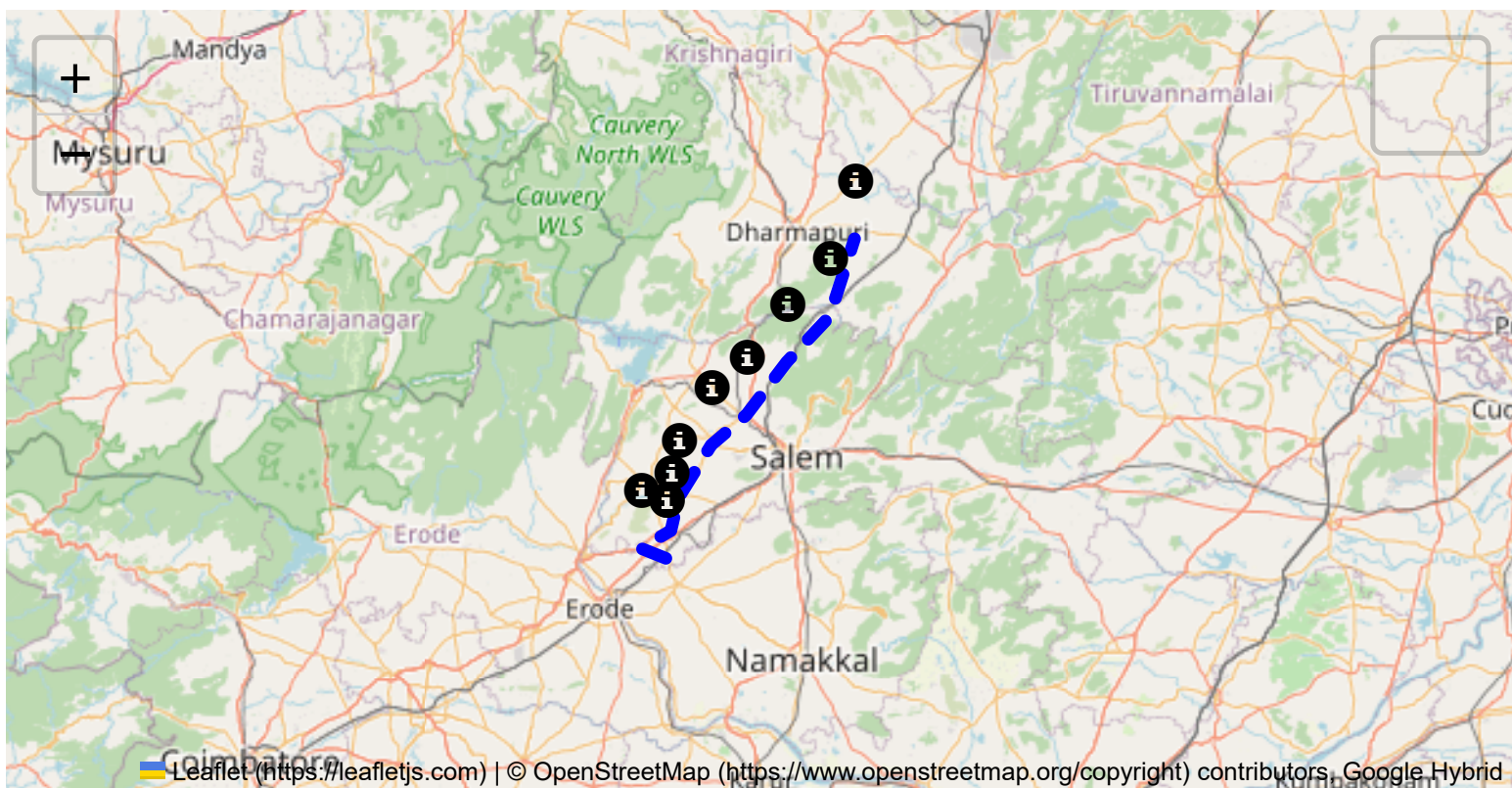
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

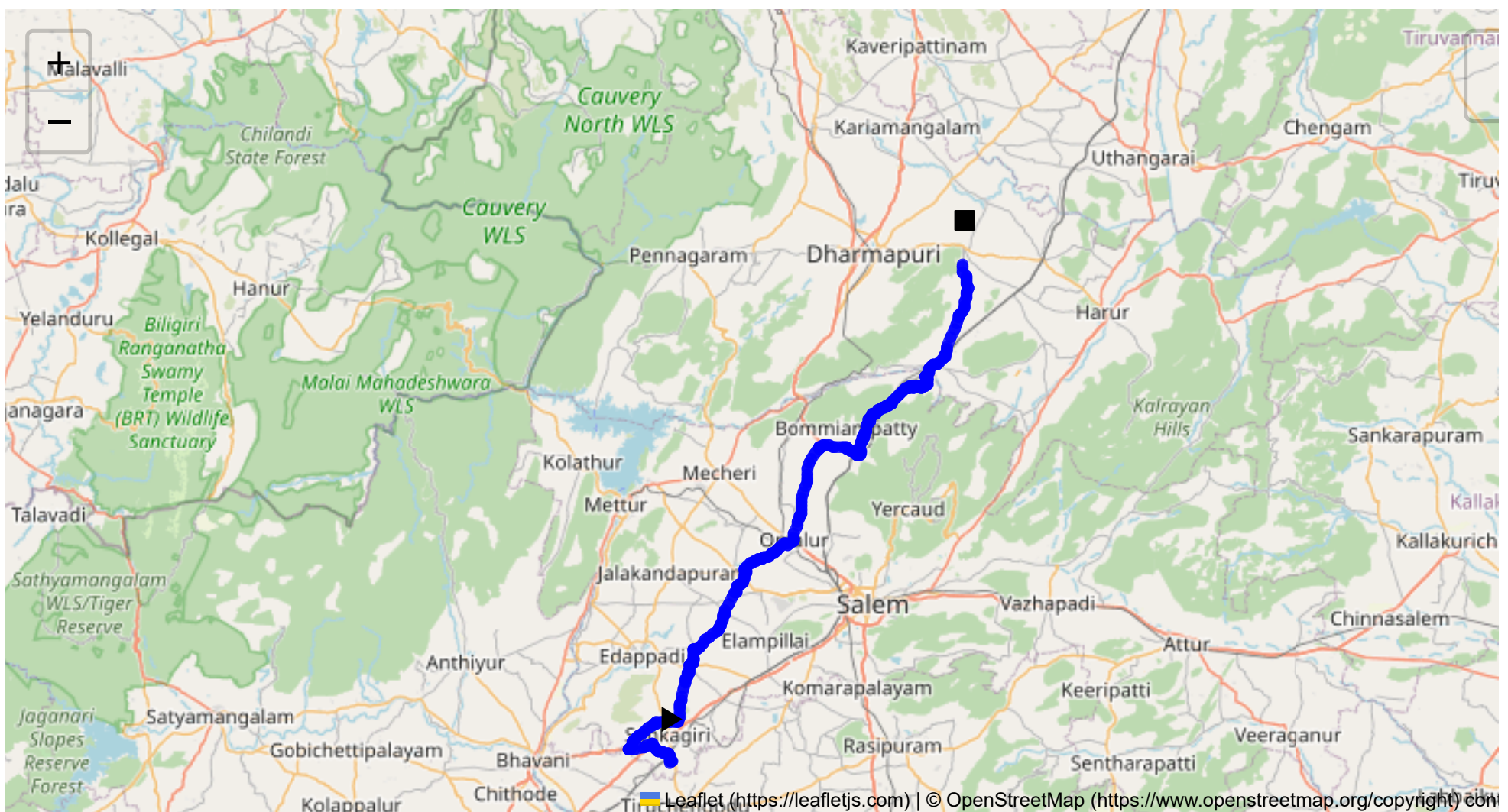
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 113.61 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.1185879, 78.284322)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route begins in Sangagiri and ends in Maniyabadi, traveling through various towns including Puthur, Padaiveedu, and Omalur. The journey spans approximately 113.61 kilometers and traverses a mix of state highways and local roads.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu typically experiences hot and dry weather with a monsoon season from June to September. During monsoons, heavy rainfall can lead to slippery roads and localized flooding. It's crucial to be cautious of reduced visibility and hydroplaning during these times.

3. Analysis of Traffic Patterns:

- Peak traffic hours often coincide with local commuting times in the morning (8-10 AM) and evening (5-8 PM). Congestion is most prominent near towns like Omalur and Salem due to local business and market activities. Expect slower movement on main roads such as the Salem-Kochi Highway.

4. Assessment of Road Quality and Infrastructure:

- Major highways like the Salem-Kochi Highway are generally well-maintained. However, some segments, especially local roads like MDR851 in rural areas, may have potholes and uneven surfaces. Road signage and lighting can be inconsistent, especially in less populated areas.

5. Suggestions for Alternative Routes for Emergencies:

- In case of road blockages or emergencies, consider rerouting via NH544 for a faster detour back towards Salem, offering better road conditions and support facilities. Another option includes SH86 towards Mettur, albeit with potentially slower travel times due to narrower lanes.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials requires adherence to strict state regulations including permissible transport hours, especially avoiding school zones and residential areas during peak hours. Necessary permits and prescribed route use are mandatory.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Historically, accidents involving heavy vehicles have occurred primarily along busy stretches near Salem. Common causes include vehicle overload and brake failures, exacerbated by sharp turns and sudden braking in congested areas.

8. Environmental Considerations and Sensitive Areas:

- The route passes through several agricultural zones. Precaution is necessary to prevent contamination from hazardous materials. Near Danishpet, additional care is required due to proximity to forested areas, which are ecologically sensitive.

9. Analysis of Communication Coverage:

- Most of the route has decent mobile network coverage, but potential dead zones exist in more rural areas, especially between Omalur and Danishpet. Satellite phones or additional communication devices are recommended for continuous connectivity.

10. Estimated Emergency Response Times:

- Urban centers like Salem can offer emergency services typically within 15-30 minutes. However, response times could extend to over an hour in more rural locations along the route, especially late at night or during adverse weather conditions.

12. Overall Summary of Risk Assessment:

- The route presents a moderate level of risk primarily due to variable road conditions, potential weather impacts, and areas of congestion. Safety measures should include pre-trip route checks,

informed driving to mitigate adverse conditions, and comprehensive emergency plans. Ensuring compliance with regulations and remaining vigilant about communication and environmental considerations will assist in reducing risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45543, 77.81740	30 KM/Hr
7	Turn	High	11.45631, 77.81668	15 KM/Hr
8	Turn	High	11.49383, 77.88442	15 KM/Hr
9	Turn	Medium	11.56083, 77.89921	30 KM/Hr
10	Turn	Medium	11.56180, 77.90019	30 KM/Hr
11	Turn	Medium	11.58254, 77.90656	30 KM/Hr
12	Turn	Medium	11.63867, 77.94841	30 KM/Hr
13	Turn	Medium	11.64217, 77.95200	30 KM/Hr
14	Turn	Medium	11.64712, 77.95118	30 KM/Hr
15	Turn	High	11.67858, 77.96834	15 KM/Hr
16	Turn	High	11.70233, 77.97869	15 KM/Hr
17	Turn	Medium	11.71394, 77.99572	30 KM/Hr
18	Turn	Medium	11.71916, 78.01117	30 KM/Hr
19	Turn	Medium	11.73787, 78.03528	30 KM/Hr
20	Turn	High	11.73651, 78.03868	15 KM/Hr
21	Turn	Medium	11.74257, 78.04708	30 KM/Hr
22	Turn	Medium	11.77998, 78.05786	30 KM/Hr
23	Turn	Medium	11.86915, 78.08382	30 KM/Hr
24	Turn	High	11.87099, 78.08358	15 KM/Hr
25	Turn	High	11.87110, 78.08383	15 KM/Hr
26	Turn	Medium	11.87072, 78.08439	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
27	Turn	Medium	11.87088, 78.08556	30 KM/Hr
28	Turn	High	11.87246, 78.08704	15 KM/Hr
29	Turn	High	11.86898, 78.12074	15 KM/Hr
30	Turn	Medium	11.86495, 78.12530	30 KM/Hr
31	Turn	High	11.85830, 78.13568	15 KM/Hr
32	Turn	Medium	11.86080, 78.13965	30 KM/Hr
33	Turn	Medium	11.86400, 78.14011	30 KM/Hr
34	Turn	Medium	11.86697, 78.13883	30 KM/Hr
35	Turn	Medium	11.88273, 78.14392	30 KM/Hr
36	Turn	Medium	11.88406, 78.14476	30 KM/Hr
37	Turn	Medium	11.88530, 78.14468	30 KM/Hr
38	Turn	Medium	11.89898, 78.14930	30 KM/Hr
39	Turn	Medium	11.90577, 78.15291	30 KM/Hr
40	Turn	High	11.90757, 78.15206	15 KM/Hr
41	Turn	Medium	11.91517, 78.15833	30 KM/Hr
42	Turn	Medium	11.92451, 78.17070	30 KM/Hr
43	Turn	Medium	11.92820, 78.18032	30 KM/Hr
44	Turn	Medium	11.92820, 78.18218	30 KM/Hr
45	Turn	Medium	11.92945, 78.18374	30 KM/Hr
46	Turn	Medium	11.93584, 78.18729	30 KM/Hr
47	Turn	Medium	11.94979, 78.20267	30 KM/Hr
48	Turn	Medium	11.95114, 78.20643	30 KM/Hr
49	Turn	Medium	11.95195, 78.20699	30 KM/Hr
50	Turn	High	11.95259, 78.20843	15 KM/Hr
51	Turn	Medium	11.95173, 78.21016	30 KM/Hr
52	Turn	Medium	11.95221, 78.21171	30 KM/Hr
53	Turn	High	11.95167, 78.21321	15 KM/Hr
54	Turn	Medium	11.95237, 78.21450	30 KM/Hr
55	Turn	High	11.95219, 78.21518	15 KM/Hr
56	Turn	Medium	11.95281, 78.21588	30 KM/Hr
57	Turn	High	11.95178, 78.22600	15 KM/Hr
58	Turn	Medium	11.96176, 78.23443	30 KM/Hr
59	Turn	High	11.96648, 78.23303	15 KM/Hr
60	Turn	Medium	11.96675, 78.23221	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
61	Turn	High	11.96695, 78.23203	15 KM/Hr
62	Turn	High	11.97321, 78.23713	15 KM/Hr
63	Turn	High	11.97761, 78.23553	15 KM/Hr
64	Turn	High	11.98244, 78.24307	15 KM/Hr
65	Turn	High	11.98207, 78.24331	15 KM/Hr
66	Turn	Medium	11.99285, 78.25796	30 KM/Hr
67	Turn	Medium	12.08633, 78.29099	30 KM/Hr
68	Turn	Medium	12.11010, 78.28440	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	clinic	Kamala Clinic	11.8723673, 78.0874195	30 km/h	Medium
30	hospital	Kamala Hospital	11.8722588, 78.0899189	30 km/h	Medium
31	hospital	GJB Hospital	11.8713998, 78.0980288	30 km/h	Medium
32	clinic	Pranav Clinic	11.8712846, 78.0998167	30 km/h	Medium
33	hospital	Bethel Hospital	11.8699472, 78.1388598	30 km/h	Medium
34	hospital	Amirtham Hospital	11.981994, 78.248038	30 km/h	Medium
35	hospital	Grs Hospital	11.9824068, 78.2481763	30 km/h	Medium
36	hospital	Government Hospital Kadathur	12.08724, 78.290259	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56180, 77.90019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58254, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77998, 78.05786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86915, 78.08382



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.87099, 78.08358



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.87110, 78.08383



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87072, 78.08439



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87088, 78.08556



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87246, 78.08704



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.86898, 78.12074



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95114, 78.20643



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97321, 78.23713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.97761, 78.23553



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98244, 78.24307



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.98207, 78.24331



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99285, 78.25796



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.08633, 78.29099



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.11010, 78.28440
